Installing and Configuring Snort on Ubuntu (Intrusion Detection System Lab)

1. Introduction

Intrusion Detection Systems (IDS) play a critical role in modern cybersecurity by monitoring network traffic and identifying suspicious or malicious activity. **Snort** is a widely used open-source IDS capable of real-time traffic analysis, packet logging, and alert generation based on defined rules.

The objective of this lab was to install Ubuntu in a virtualized environment, deploy Snort, verify its functionality, run Snort in packet sniffer mode, and create custom detection rules for ICMP and HTTP traffic. This lab demonstrates foundational blue-team skills relevant to Security Operations Center (SOC) environments.

2. Lab Environment

- **Hypervisor:** VMware Workstation Player
- **Operating System:** Ubuntu 24.04 LTS (64-bit)
- **IDS Tool:** Snort 2.9.15.1
- **Additional Tools:** Wireshark, curl
- **Network Interface:** ens33

3. Task 1 – Snort Installation Verification

3.1 Installation

Snort and Wireshark were installed using the Advanced Package Tool (APT):

sudo apt install wireshark

sudo apt install snort

During installation, the active network interface (ens33) was selected, and default HOME_NET values were accepted.

3.2 Verification

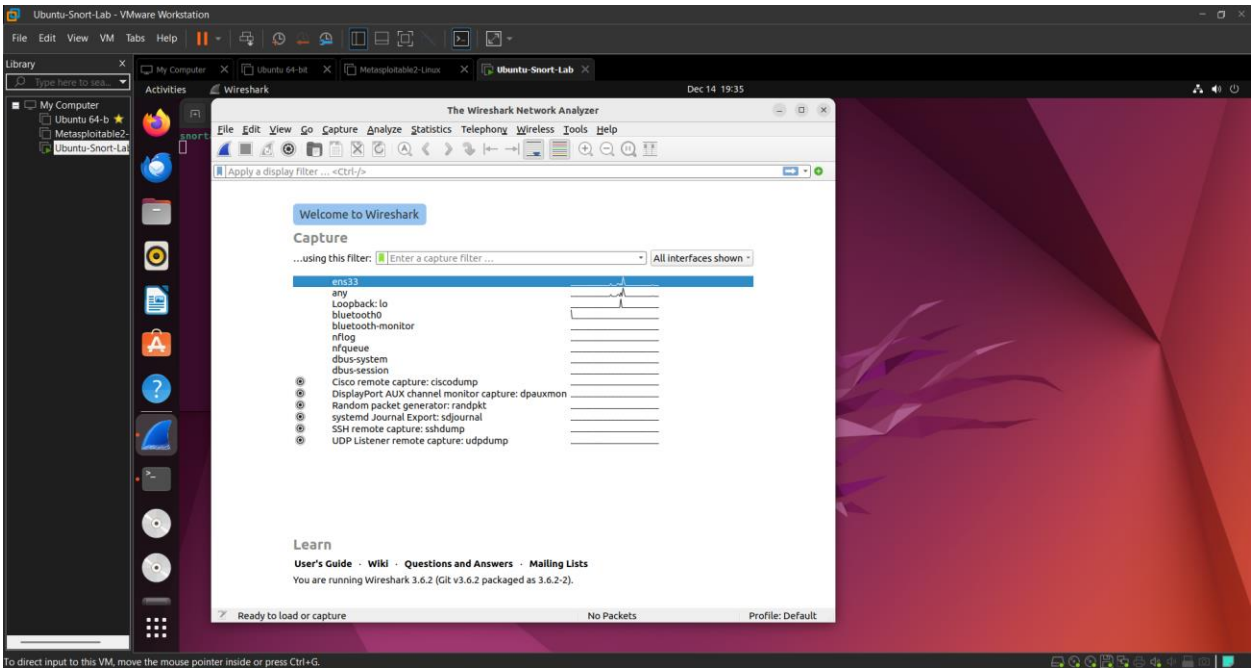
Snort installation was verified by checking the installed version and confirming the presence of the main configuration file:

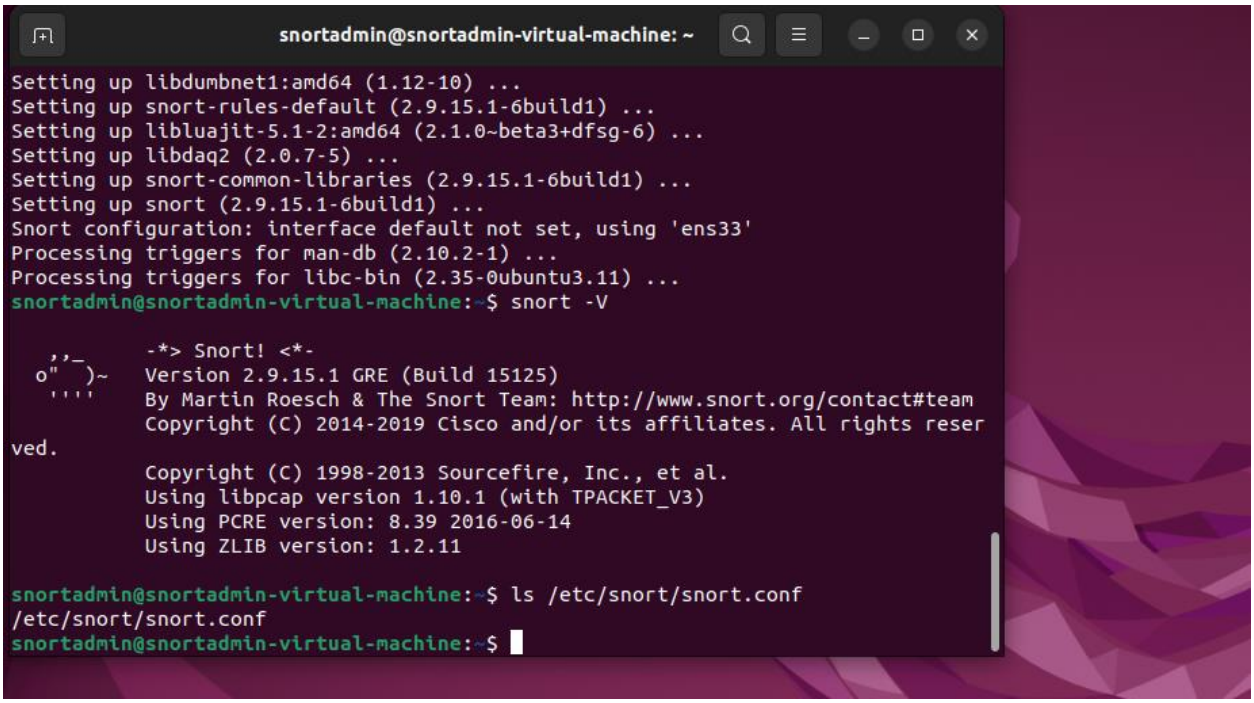
snort -V

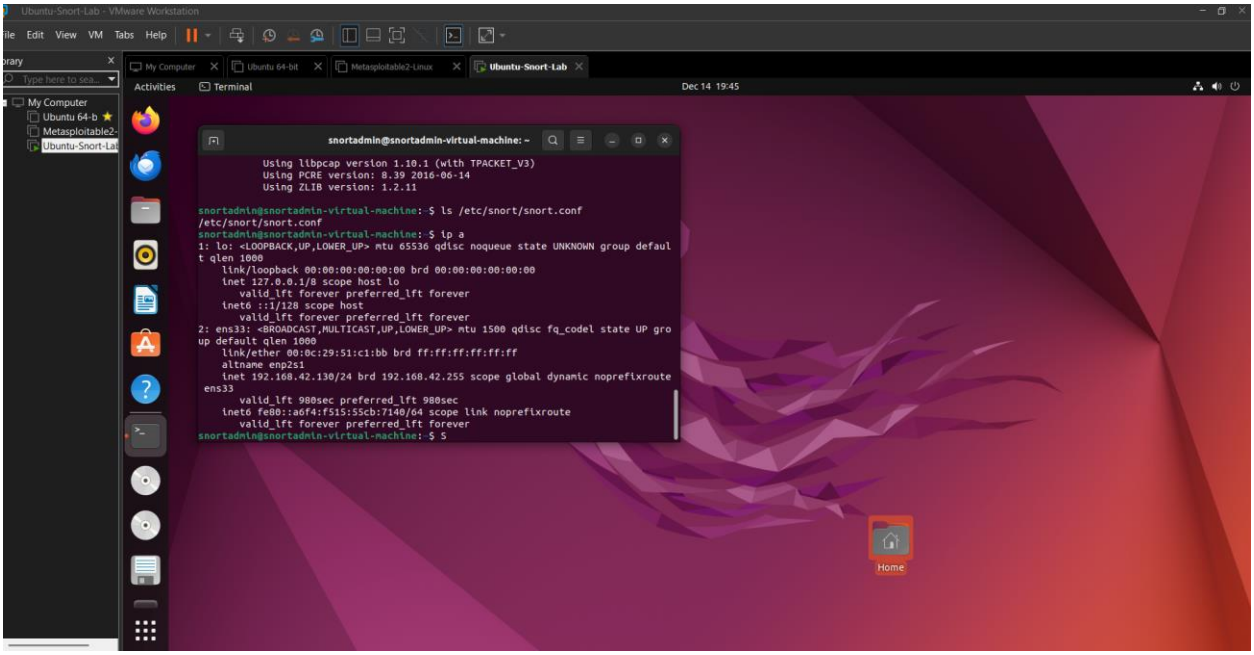
ls /etc/snort/snort.conf

Result:

- Snort version information displayed successfully
- The snort.conf configuration file was present




snortadmin@snortadmin-virtual-machine: ~
Setting up libdumbnet1:amd64 (1.12-10) ...
Setting up snort-rules-default (2.9.15.1-6build1) ...
Setting up libluajit-5.1-2:amd64 (2.1.0~beta3+dfsg-6) ...
Setting up libdaq2 (2.0.7-5) ...
Setting up snort-common-libraries (2.9.15.1-6build1) ...
Setting up snort (2.9.15.1-6build1) ...
Snort configuration: interface default not set, using 'ens33'
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
snortadmin@snortadmin-virtual-machine:~$ snort -V
-*> Snort! <*-
Version 2.9.15.1 GRE (Build 15125)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reser
ved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
snortadmin@snortadmin-virtual-machine:~$ ls /etc/snort/snort.conf
/etc/snort/snort.conf
snortadmin@snortadmin-virtual-machine:~$


Ubuntu-Snort-Lab - VMware Workstation
File Edit View VM Tabs Help
My Computer
Ubuntu 64-bit
Metasploitable2-Linux
Ubuntu-Snort-Lab
Activities
Terminal
Dec 14 19:45
snortadmin@snortadmin-virtual-machine: ~
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
snortadmin@snortadmin-virtual-machine:~$ ls /etc/snort/snort.conf
/etc/snort/snort.conf
snortadmin@snortadmin-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
link/ether 00:0c:29:51:c1:bb brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 192.168.42.130/24 brd 192.168.42.255 scope global dynamic noprefixroute
ens33
valid_lft 980sec preferred_lft 980sec
inet6 fe80::a6f4:f515:55cb:7140/64 scope link noprefixroute
valid_lft forever preferred_lft forever
snortadmin@snortadmin-virtual-machine:~$ S
Home

4. Task 2 – Running Snort in Packet Sniffer Mode

4.1 Sniffer Mode Execution

Snort was executed in packet sniffer mode to observe live network traffic:

sudo snort -v -i ens33

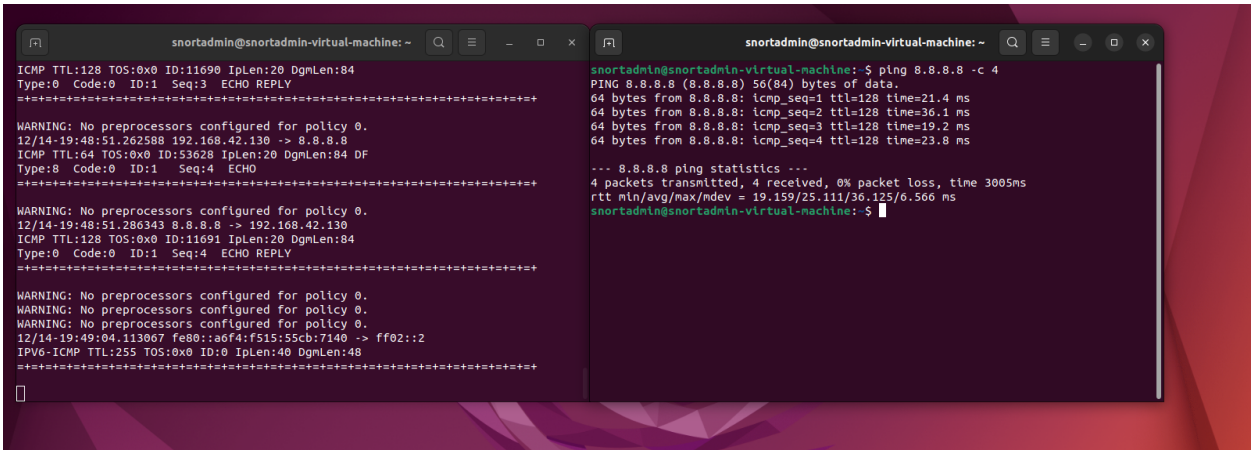
4.2 Traffic Generation

In a separate terminal, ICMP traffic was generated:

ping 8.8.8.8 -c 4

4.3 Observation

Snort successfully captured and displayed ICMP packet headers in real time, confirming that it was actively monitoring network traffic.



5. Task 3 – ICMP Detection Rule Creation

5.1 Custom Rule Creation

A custom Snort rule was created to detect ICMP ping traffic:

sudo nano /etc/snort/rules/local.rules

Rule added:

alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

5.2 Rule Execution

Snort was restarted using the configuration file:

sudo snort -A console -c /etc/snort/snort.conf -i ens33

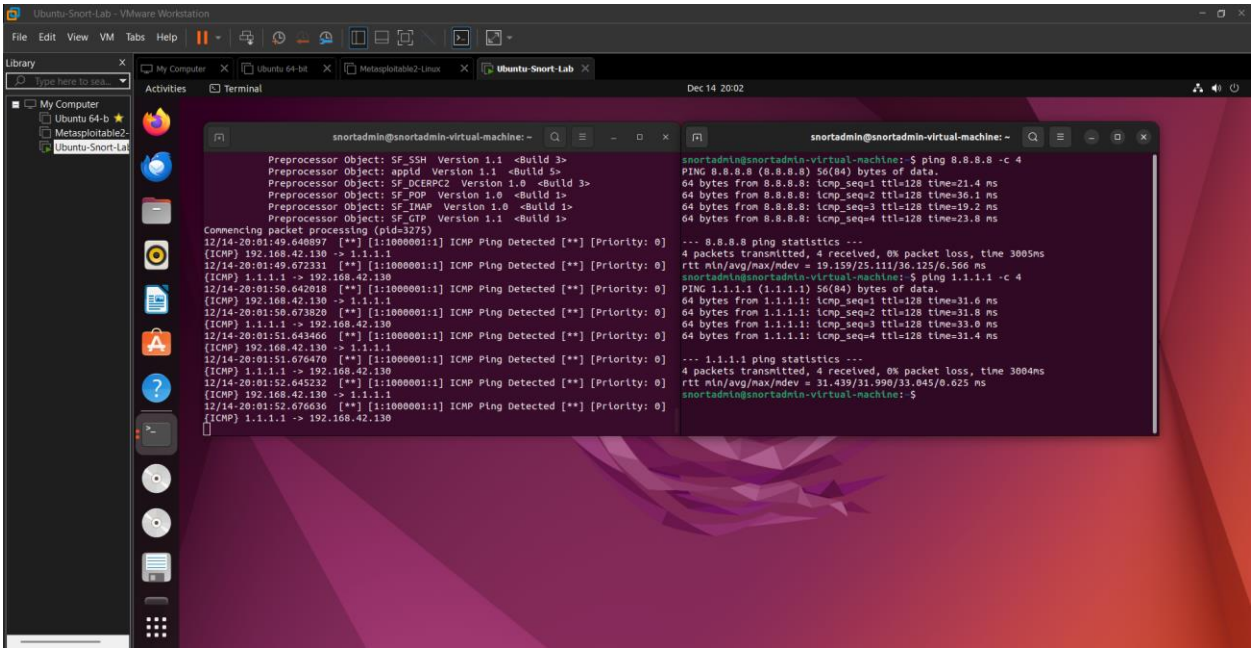
ICMP traffic was generated:

ping 1.1.1.1 -c 4

5.3 Result

Snort generated alerts indicating detection of ICMP traffic:

[**] [1:1000001:1] ICMP Ping Detected [**]



6. Task 4 – HTTP Traffic Detection Rule

6.1 Rule Design Consideration

Snort 2.9 has limitations when inspecting encrypted or browser-optimized HTTP traffic. Therefore, HTTP detection was implemented by monitoring TCP traffic destined for port 80, which reliably indicates plaintext HTTP communication.

6.2 HTTP Rule Creation

The local rules file was updated to include an HTTP detection rule:

alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000002; rev:3;)

6.3 Rule Execution

Snort was restarted to load the updated rules:

sudo snort -A console -c /etc/snort/snort.conf -i ens33

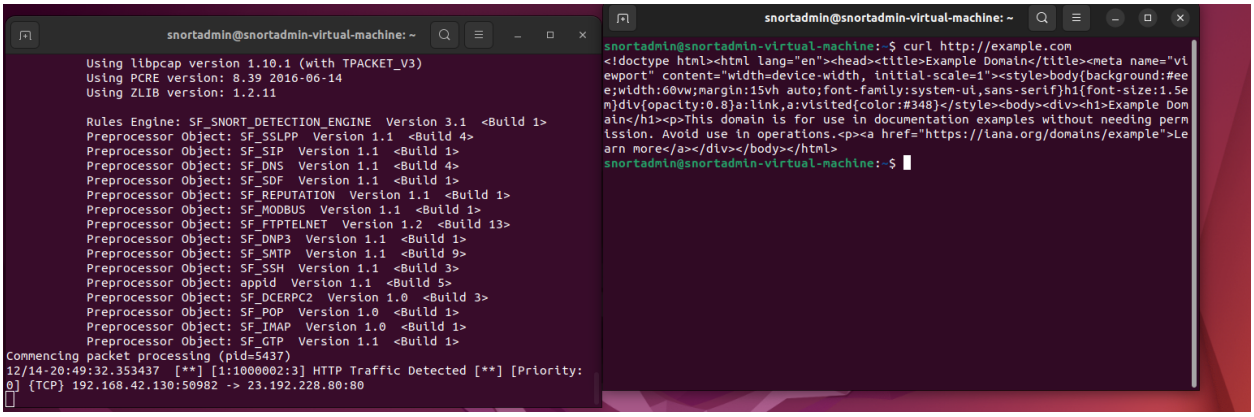
HTTP traffic was generated using curl:

curl http://example.com

6.4 Result

Snort successfully detected HTTP traffic on port 80 and generated alerts:

[**] [1:1000002:3] HTTP Traffic Detected [**]



7. Conclusion

This lab successfully demonstrated the installation, configuration, and operation of Snort as an Intrusion Detection System on Ubuntu. Snort was validated in packet sniffer mode, and custom rules were created to detect ICMP and HTTP traffic. The exercise highlighted practical IDS concepts, including rule creation, alert generation, and the importance of restarting services after configuration changes.

The skills demonstrated in this lab are directly applicable to real-world SOC environments, where IDS tools are used to monitor network activity, detect threats, and support incident response.

8. Real-World Application

In enterprise environments, Snort rules are continuously tuned to reduce false positives and improve detection accuracy. Alerts generated by Snort are typically forwarded to a Security Information and Event Management (SIEM) system for correlation and analysis. Understanding how to create and validate IDS rules is essential for effective network defense and threat monitoring.